CHAPTER 10

Permutations and Combinations

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10.1 The basic counting rules

Theorem 10.1 (Sum rule)**.** If a task can be done in n_1 ways or, *exclusively*, in n_2 ways, it can be done in $n_1 + n_2$ ways. Set version: disjoint sets satisfy $|A \cup B| = |A| + |B|$.

Theorem 10.2 (Product rule)**.** If a procedure consists of two successive independent choices, with n_1 options for the first and n_2 for the second, there are $n_1 n_2$ outcomes. Set version: $|A \times B| = |A|\,|B|$.

Example 10.3. A password of 3 letters (26 choices each) followed by 2 digits: $26^3 \cdot 10^2 =$ 1,757,600. A menu offering 4 starters or 6 mains as a single dish: $4 + 6 = 10$ choices. Tree diagrams display such combined choices as branches; the leaves count the outcomes.

10.2 Permutations

Definition 10.4. A *permutation* of n distinct objects is an ordering of all of them. The number of permutations is $n! = n(n-1)(n-2)\cdots 2 \cdot 1$, with $0! = 1$.

Theorem 10.5 (k-permutations)**.** The number of ordered arrangements of k objects chosen from n distinct objects is

$$P(n,k) = n(n-1)\cdots(n-k+1) = \frac{n!}{(n-k)!}.$$

Example 10.6 (Sprint podium)**.** Nine athletes run a sprint; podium orders (gold, silver, bronze) number $P(9,3) = 9 \cdot 8 \cdot 7 = 504$.

Theorem 10.7 (Permutations with repeated objects)**.** The number of distinct arrangements of n objects of which n_1 are alike of type 1, …, n_k alike of type k ($n_1 + \cdots + n_k = n$) is

$$\frac{n!}{n_1!\,n_2!\cdots n_k!}.$$

Example 10.8. "MISSISSIPPI" has $\frac{11!}{1!\,4!\,4!\,2!} = 34650$ distinct arrangements.

Theorem 10.9 (Circular permutations)**.** n distinct people can be seated around a round table (rotations considered identical) in $(n-1)!$ ways.

10.3 Combinations

Definition 10.10. A *combination* of k objects from n is an unordered selection. Their number is the *binomial coefficient*

$$C(n,k) = \binom{n}{k} = \frac{n!}{k!\,(n-k)!} = \frac{P(n,k)}{k!}.$$

Example 10.11. Choosing a committee of 5 from 30 students: $\binom{30}{5} = 142506$. Note order is irrelevant — choosing {Amal, Badr} equals choosing {Badr, Amal}.

Theorem 10.12 (Symmetry and Pascal's identity)**.**

$$\binom{n}{k} = \binom{n}{n-k}, \qquad \binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}.$$

Proof. Symmetry: choosing the k selected is the same as choosing the $n-k$ rejected. Pascal: fix one element x of an $(n+1)$-set; the k-subsets split into those containing x ($\binom{n}{k-1}$ of them) and those not ($\binom{n}{k}$). □

Pascal's identity generates *Pascal's triangle*, whose row n lists $\binom{n}{0}, \dots, \binom{n}{n}$:

$$\begin{array}{ccccccccc}
 & & & & 1 & & & & \\
 & & & 1 & & 1 & & & \\
 & & 1 & & 2 & & 1 & & \\
 & 1 & & 3 & & 3 & & 1 & \\
1 & & 4 & & 6 & & 4 & & 1
\end{array}$$

Row sums give $\sum_k \binom{n}{k} = 2^n$ (each subset of an n-set has some size k — recounting the power-set theorem of Chapter 6).

10.4 The binomial theorem

Theorem 10.13 (Binomial theorem)**.** For all x, y and $n \in \mathbb{N}$:

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k.$$

Proof. Expanding $(x+y)(x+y)\cdots(x+y)$, each term arises by choosing y from k of the n factors and x from the rest; there are $\binom{n}{k}$ such choices, giving the coefficient of $x^{n-k}y^k$. ☐

10.5 Combinations with repetition

Theorem 10.14 (Stars and bars)**.** The number of multisets of size k from n types — equivalently the number of solutions of $x_1 + \cdots + x_n = k$ in nonnegative integers — is

$$\binom{n+k-1}{k}.$$

Proof. Encode a selection as k stars separated by $n-1$ bars; a word with $n+k-1$ symbols is determined by the positions of the k stars. ☐

10.5.1 The four counting problems

Select k from n	order matters	order irrelevant
without repetition	$P(n,k) = \dfrac{n!}{(n-k)!}$	$\binom{n}{k}$
with repetition	n^k	$\binom{n+k-1}{k}$

10.6 Multinomial coefficients

Theorem 10.15 (Multinomial theorem)**.**

$$(x_1+\cdots+x_m)^n = \sum_{n_1+\cdots+n_m=n} \binom{n}{n_1, n_2, \ldots, n_m} x_1^{n_1}\cdots x_m^{n_m}, \qquad \binom{n}{n_1,\ldots,n_m} = \frac{n!}{n_1!\cdots n_m!},$$

the multinomial coefficient counting the ways to partition n labelled objects into groups of sizes $n_1, \ldots, n_m$ (cf. Theorem 10.7).

10.7 The pigeonhole principle

Theorem 10.16 (Pigeonhole principle)**.** If $n+1$ or more objects are placed into n boxes, some box contains at least two objects. Generalised form: with N objects in n boxes, some box contains at least $\lceil N/n \rceil$ objects.

Example 10.17. Among any 13 people, two share a birth month. Among 100 people, at least $\lceil 100/12 \rceil = 9$ share one. Subtler: among any $n+1$ integers chosen from $\{1, \ldots, 2n\}$, two are consecutive — pair the numbers $\{1,2\}, \{3,4\}, \ldots, \{2n-1, 2n\}$ and apply pigeonhole.

10.8 Inclusion–exclusion

Theorem 10.18. For finite sets $A_1, \ldots, A_n$:

$$\Big|\bigcup_i A_i\Big| = \sum_i |A_i| - \sum_{i<j} |A_i \cap A_j| + \sum_{i<j<k} |A_i \cap A_j \cap A_k| - \cdots + (-1)^{n+1} |A_1 \cap \cdots \cap A_n|.$$

Example 10.19 (Derangements)**.** A *derangement* is a permutation with no fixed point. Let A_i be the permutations fixing i; inclusion–exclusion gives

$$D_n = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!} \quad \xrightarrow[n\to\infty]{} \quad \frac{n!}{e}.$$

So if n guests grab random hats, the chance nobody gets their own tends to $1/e \approx 0.3679$.

10.9 Two classical identities

Theorem 10.20 (Vandermonde)**.** $\displaystyle\binom{m+n}{r} = \sum_{k=0}^{r} \binom{m}{k}\binom{n}{r-k}$ — choose r people from m women and n men by cases on how many women.

Theorem 10.21 (Hockey stick)**.** $\displaystyle\sum_{i=r}^{n} \binom{i}{r} = \binom{n+1}{r+1}$ — classify the $(r+1)$-subsets of $\{0, \ldots, n\}$ by their largest element.

Example 10.22 (Poker hands)**.** From a 52-card deck, $\binom{52}{5} = 2598960$ five-card hands. Full houses: choose the triple rank, its suits, the pair rank, its suits: $13 \cdot \binom{4}{3} \cdot 12 \cdot \binom{4}{2} = 3744$.

10.10 Strategy: choosing the right formula

Ask in order: (1) Am I *selecting* or *arranging* (does order matter)? (2) Is repetition allowed? — this places you in the four-problem table. (3) Are objects distinguishable? If types repeat, use multiset permutations or stars and bars. (4) "At least/at most" constraints often yield to complement counting or inclusion–exclusion; existence claims to pigeonhole.

10.11 Summary

Sum and product rules compose all elementary counts. Order + no repetition: $P(n,k)$; no order: $\binom{n}{k}$; with repetition: n^k and $\binom{n+k-1}{k}$. Pascal's identity builds the triangle; the binomial and multinomial theorems expand powers; pigeonhole guarantees collisions; inclusion–exclusion corrects overcounting — derangements being its signature application.

Exercises

Exercise 10.1 EASY

License plates consist of 2 letters followed by 3 digits. How many are possible? How many with no repeated symbol?

Exercise 10.2 EASY

Compute $P(7,3)$, $\binom{7}{3}$, and the number of binary strings of length 7 containing exactly three 1s.

Exercise 10.3 EASY

How many distinct arrangements has the word "ABRACADABRA"?

Exercise 10.4 MEDIUM

A committee of 6 is chosen from 10 men and 8 women. How many committees contain (a) exactly 3 women; (b) at least one woman?

Exercise 10.5 MEDIUM

How many solutions in nonnegative integers has $x+y+z=11$? How many with $x \geq 2$?

Exercise 10.6 MEDIUM

Find the coefficient of x^5 in $(2x-3)^8$.

Exercise 10.7 MEDIUM

In how many ways can 8 people sit around a round table? What if two specific people insist on sitting together?

Exercise 10.8 MEDIUM

Show that among any $n+1$ integers from $\{1,2,\ldots,2n\}$, one divides another. *Hint: write each as $2^a m$ with m odd, and apply pigeonhole to the odd parts.*

Exercise 10.9 MEDIUM

How many integers in $\{1,\ldots,1000\}$ are divisible by 2, 3 or 5?

Exercise 10.10 HARD

Prove Vandermonde's identity (Theorem 10.20) by computing the coefficient of x^r in $(1+x)^m(1+x)^n = (1+x)^{m+n}$.

Exercise 10.11 HARD

Four guests check four hats and receive them back at random. Using Example 10.19, count the outcomes where nobody receives their own hat, and the probability of that event.

Exercise 10.12 HARD

How many surjections are there from a 5-element set onto a 3-element set? *Hint: inclusion–exclusion over the missed elements:* $\sum_j (-1)^j \binom{3}{j}(3-j)^5$.

Solutions to the Exercises

Solution 10.1. $26^2 \cdot 10^3 = 676000$. Without repetition: $26 \cdot 25 \cdot 10 \cdot 9 \cdot 8 = 468000$. ■

Solution 10.2. $P(7,3) = 7 \cdot 6 \cdot 5 = 210$; $\binom{7}{3} = 35$; a string is determined by the positions of its three 1s: $\binom{7}{3} = 35$. ■

Solution 10.3. Letters: A×5, B×2, R×2, C, D — total 11: $\frac{11!}{5!\,2!\,2!\,1!\,1!} = 83160$. ■

Solution 10.4. (a) $\binom{8}{3}\binom{10}{3} = 56 \cdot 120 = 6720$. (b) Total minus all-male: $\binom{18}{6} - \binom{10}{6} = 18564 - 210 = 18354$. ■

Solution 10.5. Stars and bars ($n = 3$, $k = 11$): $\binom{13}{2} = 78$. With $x \geq 2$, substitute $x' = x - 2$: $x' + y + z = 9$, giving $\binom{11}{2} = 55$. ■

Solution 10.6. General term $\binom{8}{k}(2x)^{8-k}(-3)^k$; x^5 needs $k = 3$: $\binom{8}{3}2^5(-3)^3 = 56 \cdot 32 \cdot (-27) = -48384$. ■

Solution 10.7. $(8-1)! = 5040$. Glue the pair into one block: $(7-1)! = 720$ circular orders, times 2 internal arrangements: 1440. ■

Solution 10.8. Write each chosen integer as $2^a m$ with odd part $m \in \{1, 3, \ldots, 2n-1\}$ — only n possible odd parts for $n+1$ numbers, so two share the same m: say $2^a m$ and $2^b m$ with $a < b$. Then $2^a m \mid 2^b m$. ■

Solution 10.9. With D_k = multiples of k: $|D_2| = 500$, $|D_3| = 333$, $|D_5| = 200$, $|D_6| = 166$, $|D_{10}| = 100$, $|D_{15}| = 66$, $|D_{30}| = 33$. Inclusion–exclusion: $500 + 333 + 200 - 166 - 100 - 66 + 33 = 734$. ■

Solution 10.10. The coefficient of x^r in $(1+x)^{m+n}$ is $\binom{m+n}{r}$. In the product $(1+x)^m(1+x)^n = \left(\sum_k \binom{m}{k}x^k\right)\left(\sum_j \binom{n}{j}x^j\right)$, the x^r terms pair $k + j = r$: coefficient $\sum_{k=0}^{r} \binom{m}{k}\binom{n}{r-k}$. Equate. ■

Solution 10.11. $D_4 = 4!\left(1 - 1 + \frac{1}{2} - \frac{1}{6} + \frac{1}{24}\right) = 24 \cdot \frac{9}{24} = 9$ derangements out of $4! = 24$ outcomes: probability $\frac{9}{24} = \frac{3}{8} = 0.375$ (already close to $1/e$). ■

Solution 10.12. $\sum_{j=0}^{3}(-1)^j\binom{3}{j}(3-j)^5 = 3^5 - 3 \cdot 2^5 + 3 \cdot 1^5 - 0 = 243 - 96 + 3 = 150$. ■